

Assignment 1

1. Write a function that takes a string and returns its reverse **without using slicing** (`[::-1]`).

Example:

Input: hello

Output: olleh

2. Write a function `count_even(lst)` that takes a list of integers and returns the number of even numbers.

Example:

Input: [1, 2, 3, 4, 6]

Output: 3

3. Write a function that takes a list of numbers and returns the largest number **without using** `max()`.

4. Write a function that counts the number of vowels in a given string.

Example

Input: "Artificial Intelligence"

Output: 10

5. Write a function that removes duplicate elements from a list while preserving the original order.

Example

Input:

[1,2,2,3,4,4,5]

Output:

[1,2,3,4,5]

6. Write a program that prints all prime numbers from **1 to N**.

Example

Input:

20

Output:

2 3 5 7 11 13 17 19

7. Write a function that returns a dictionary containing the frequency of each character in a string.

Example

Input:

"banana"

Output:

{'b':1,'a':3,'n':2}

8. Write a function that takes two matrices (lists of lists) of the same size and returns their sum.

Example

A = [[1,2],[3,4]]

B = [[5,6],[7,8]]

Output:

[[6,8],[10,12]]

9. Write a function that takes a sentence and returns the frequency of each word.

Example

Input:

"python is easy python is fun"

Output:

```
{  
'python':2,  
'is':2,  
'easy':1,  
'fun':1  
}
```

10. Write a program using nested loops to print the following pattern for N = 5.

```
*  
  
**  
  
***  
  
****  
  
*****  
  
****  
  
***  
  
**  
  
*
```